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Assessing Fairness in Decision-Making Algorithms for Contextual Bandit Problems

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Chapter 1

Introduction

1.1 Background

Uncertainty is a fundamental feature of our world. Faced with it, humans rely on various strategies to make decisions: flipping a coin, following their instincts, adhering to social norms, or acting in self-interest. However, as shown by Kahneman's work on cognitive biases and dual system thinking[Kahneman, 2011], human decision making is often far from rational, instead shaped by intuition, heuristics, emotions, and bounded reasoning. Although rationality would suggest evaluating the consequences of actions based on probabilistic reasoning, a perspective dating back to Pascal's pioneering work on decision under uncertainty[Le Gall, 2023], this ideal is rarely achieved in practice.

In contrast, algorithmic decision-making seems more consistent and rational. Algorithms follow formal procedures based on probability and decision theory[Berger, 1985], using models to optimize outcomes under uncertainty. For example, multi-armed bandit (MAB) algorithms formalize the trade-off between exploration and exploitation[Lattimore and Szepesvári, 2020, Auer et al., 2002a] and are widely applied in areas such as online recommendation[Li et al., 2010], clinical trials[Villar et al., 2015], and adaptive experimentation.

However, algorithmic decisions are not necessarily fair. They can replicate or even exacerbate existing social biases, especially when trained on historical data or deployed in dynamic environments where feedback loops emerge[Binns, 2018, Dwork et al., 2012]. This is well-documented in bandit settings, where algorithms may consistently under-serve certain groups based on early outcomes, leading to long-term disparities[Joseph et al., 2016, Liu and Simchi-Levi, 2017].

Fairness remains a fundamentally human notion, reflecting our values around justice, equal treatment, and individual dignity[Rawls, 1971]. While many mathematical definitions of fairness exist [Caton et al., 2022, Rabonato and Berton, 2024], the central difficulty lies not in formalizing fairness, but in reconciling competing definitions and selecting the one most appropriate for the context. Asking an algorithm to be fair thus requires a deliberate modeling effort, normative choices, and a clear awareness of ethical trade-offs.

Machine learning algorithms optimize for accuracy or utility based on the data they are given and the objectives they are designed to achieve. However, these objectives rarely include fairness constraints unless they are explicitly specified. Furthermore, algorithmic systems lack contextual understanding or moral judgment; they operate solely based on statistical patterns, not normative values. The importance of addressing these limitations was underscored in a 2018 report by the Council of Europe, titled Discrimination, Artificial Intelligence, and Algorithmic Decision-Making[Zuiderveen Borgesius et al., 2018]. Prepared for the Council's Anti-Discrimination Department, the report highlights how AI, despite its potential to enhance efficiency, healthcare, and economic development, can also lead to discriminatory

outcomes. These arise, for example, when systems learn from biased historical data or reflect societal prejudices. The report uses the term "discrimination" to refer specifically to harmful or unlawful differentiation, such as that based on gender, skin color, or ethnic origin, and calls for safeguards to prevent such outcomes in algorithmic systems.

In many real-world applications, such as personalized advertising, healthcare resource allocation, and recommendation systems, decisions must be made sequentially under uncertainty and with limited feedback. The *contextual multi-armed bandit* (CMAB) framework offers a principled way to model such problems, where at each round, the agent observes a context and selects an action (or "arm") to maximize cumulative reward over time [Lattimore and Szepesvári, 2020, Slivkins, 2019].

While the standard CMAB framework focuses on optimizing performance through exploration-exploitation trade-offs, it does not explicitly address the fairness of the decisions made, particularly when contexts encode sensitive attributes (e.g., gender, ethnicity, socio-economic background). This has raised significant concerns, as data-driven decision-making systems risk perpetuating or amplifying social biases if fairness is not properly considered [Barocas et al., 2023].

Recent research has proposed definitions and algorithmic solutions to incorporate fairness into CMAB settings[Rabonato and Berton, 2024]. Notions such as group fairness, which aims to ensure equitable treatment across protected groups, and individual fairness, which requires that similar individuals receive similar outcomes, have been extended to the sequential decision-making context, including MABs[Joseph et al., 2016, Liu and Simchi-Levi, 2017]. These adaptations seek to reconcile the need for exploration and exploitation with fairness constraints over time. However, other formalizations of fairness, such as counterfactual fairness[Kusner et al., 2017] or fairness through unawareness [Dwork et al., 2012] have not yet been systematically studied in bandit settings. Exploring how these fairness notions interact with sequential learning dynamics remains an open and promising direction for future research. For example, the FairUCB algorithm [Joseph et al., 2016] constrains action selection to guarantee equitable exposure across groups while maintaining regret. Other works propose fairness-aware reward modeling, constrained optimization techniques, or regret decomposition by subpopulations [Jabbari et al., 2017].

Nevertheless, balancing fairness and performance in CMAB remains a challenging problem. Existing approaches, such as those based on group or individual fairness
[Joseph et al., 2016, Liu and Simchi-Levi, 2017], often rely on restrictive assumptions, such as access to protected attributes, predefined group structures, or fixed reward distributions. These models typically formalize fairness through hard constraints (e.g., fairness regret bounds or demographic parity), which can limit learning efficiency, especially in non-stationary or high-dimensional settings. Moreover, many of these frameworks assume a static notion of fairness over time, neglecting the dynamic and cumulative impact of sequential decisions. This is problematic in real-world applications, where exploration policies may create unfair short-term outcomes even if fairness is achieved asymptotically. Additionally, other fairness notions, such as counterfactual or causal fairness[Kusner et al., 2017] have not been integrated into bandit frameworks, leaving open the question of how to balance long-term utility with more nuanced ethical considerations. Therefore, designing CMAB algorithms that offer both fairness and competitive performance under realistic assumptions remains an open and important research direction. Key trade-offs involve long-term fairness versus short-term reward maximization, computational complexity, and robustness to unknown or evolving context distributions. Furthermore, the lack of standardized evaluation benchmarks for fairness in CMAB hinders systematic comparison of existing algorithms.

1.2 The Decision-Making Process

1.2.1 Algorithmic decision-making

Probability theory provides a structured way to represent uncertainty in a precise, mathematical form [Jaynes, 2003]. Decision theory builds on this by offering methods to make optimal choices under uncertainty[Berger, 1985, Savage, 1954]. In practice, algorithmic decision-making often follows formal rules, typically by framing decisions as optimization problems where the goal is to maximize rewards or minimize losses. Many algorithms account for uncertainty by using probabilities and expected outcomes, drawing on models such as Bayesian decision theory, Markov decision processes, or MABs[Russell and Norvig, 2010, Lattimore and Szepesvári, 2020].

Unlike humans, algorithms do not rely on intuition or instinct. They operate based on consistent and logical rules. However, poor algorithmic decisions can still occur, not because the algorithm is irrational, but because it relies on biased data or flawed assumptions[Barocas and Selbst, 2016]. These algorithmic biases differ from human biases in how they arise. Human biases often stem from personal experiences, cultural context, or unconscious prejudices[Kahneman, 2011]. In contrast, algorithmic biases typically originate from the training data or the way the system was designed. For example, if an algorithm is trained on historical data that reflects social inequalities, it may replicate or even amplify those inequalities in its decisions[Mehrabi et al., 2021]. The key difference is that, while human biases are often subtle and difficult to detect, algorithmic biases can be identified and corrected more systematically, through data audits, careful modeling, or fairness constraints[Mitchell et al., 2021]. Still, the consequences can be similar: whether decisions are made by humans or machines, relying on biased inputs can reinforce existing disparities.

1.2.2 Collective algorithmic decision-making

Researchers have investigated whether combining different models can enhance decision-making. This exploration led to the development of ensemble methods. These methods involve combining multiple machine learning models to achieve better predictive performance than could be obtained from any individual model alone. Common examples include bagging, boosting, and stacking. The main objective of ensemble learning is to find an optimal performance through the combination of multiple classifiers. In ensemble learning a single or multiple algorithms is deployed to generate different base classifiers. These base classifiers are strategically combined together through a combination method of decision making in classifying new data instances. Since an ensemble is a combination of multiple methods, assessing the prediction of an ensemble requires a lot of computation compared to that of a single model.

Bagging can help reduce variance in reward predictions by averaging outputs over multiple models trained on bootstrapped samples of context/reward pairs. This is particularly useful in noisy or non-stationary environments. *Boosting* improves prediction performance by sequentially focusing on hard-to-predict instances and can enhance the model's expressiveness in complex feature spaces, especially when combined with techniques to estimate uncertainty. *Stacking* allows for the combination of diverse models (e.g., specialized for different user groups or context regions).

In the classical multi-armed bandit (MAB) setting, an agent repeatedly selects one action (or "arm") among a finite set and receives a stochastic reward, with the goal of maximizing cumulative reward over time. The contextual multi-armed bandit (CMAB) extends this framework by incorporating side information ("context") available at each decision point, allowing the agent to condition its choices on observed features. To effectively integrate these ensemble methods into CMAB algorithms, one must pair them with exploration-aware strategies such as Upper Confidence Bounds (UCB) or Thompson Sampling. For instance, reward predictions from an ensemble can be used in a UCB framework by incorporating prediction variance to guide exploration[Lu et al., 2010, Bastani et al., 2020, Riquelme et al., 2018].

1.3 The Importance of Fairness in Decision-Making

Fairness in computer science has become an important topic, especially as numerous reports have highlighted concerns about algorithmic bias[Barocas and Selbst, 2016, Zuiderveen Borgesius et al., 2018, O'Neil, 2016, com, 2020]. Some experts argue that algorithms are so deeply affected by biases that their decisions cannot be considered fair or reliable. Consequently, they recommend rigorously evaluating AI decision-making systems with standards similar to those applied in pharmaceutical drug testing before approval for widespread use. Algorithmic bias occurs when systematic errors in machine learning algorithms produce unfair or discriminatory outcomes, often reflecting or reinforcing existing socioeconomic, ethnic, and gender biases. In this case, algorithmic bias is not caused by the algorithm itself, but by how the data science team collects and codes the training data. For example, United States courts use the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) tool to assess defendants' recidivism risk; but in this risk assessment, black defendants were twice as likely as white defendants to be misclassified as being a higher risk of violent recidivism [Berk et al., 2021]. Another example is the lack of fairness in the electronic health records of patients due to undergo coronary artery bypass surgery [Zhang et al., 2024].

It is very difficult to estimate the degree of fairness because we want to achieve equal opportunities for all. However, the parameters we aim to evaluate could already display uneven distributions at the beginning, which introduces potential biases and must be accounted for when interpreting the fairness and performance of the algorithms. Those different distributions will therefore have to be taken into account both before and after comparisons between the different groups. For instance, consider the case of atrial fibrillation, a condition that is more prevalent in men than in women, and more common among White individuals than among black people and those of Asian descent. Fair treatment will have to take into account the differences in prevalence before and after the decision is made by the algorithms. It is also important to consider that a truly fair algorithm may need to incorporate elements of *positive discrimination*, especially to correct for historical or structural imbalances in the data.

1.3.1 The Importance of Fairness Across Domains

Fairness is a fundamental principle that underpins ethical behavior and trust in many domains where algorithmic decisions are increasingly used. In strategic environments such as games, fairness ensures balanced opportunities among players and prevents systemic advantages that undermine the integrity of competition. In medicine, fairness is crucial to guarantee equitable access to diagnosis, treatment, and preventive care across different demographic groups; biased models can lead to life-threatening disparities in health outcomes. In the financial sector, fairness is essential to avoid discriminatory lending practices or investment strategies that exclude vulnerable populations. Similarly, in the judicial system, fairness is not only a legal obligation but also a prerequisite for social trust: Algorithmic tools used to assess risk of recidivism or inform sentencing must be carefully audited to prevent reinforcing existing biases. These examples underscore that fairness is not merely a technical constraint but a social imperative that must be explicitly considered when deploying learning algorithms in real-world applications.

1.3.2 Temporal Dynamics of Fairness

Fairness is not a static property; it can evolve over time as learning algorithms adapt to data and user feedback. In online decision-making settings, early rounds may exhibit biased behavior due to limited information, while later rounds may become fairer as the algorithm explores more diverse options and learns from a broader set of contexts. However, the opposite can also occur: feedback loops or data drift may gradually increase unfairness if not properly monitored. Therefore, it is essential to evaluate fairness not only as a global metric but also dynamically over time, using temporal analyses and visualizations to detect potential disparities as they emerge.

1.3.3 Possible models to assess fairness

Designing machine learning models that provide equitable outcomes is challenging, considering that fairness can be defined in different ways. Fairness has a number of different definitions and measures, such as demographic parity, equalized odds, and individual fairness [Dwork et al., 2012] [Barocas et al., 2019]. This means that choosing the right model depends on how fairness is defined, what kind of data is available, and the context in which the model will be used.

Simple and interpretable models like decision trees and logistic regression are typically preferred when understanding how the model makes decisions is important, whereas more complex ensemble methods like random forests offer higher predictive power at the cost of interpretability. These models enable us to see if there is any pattern of discrimination or bias. This is why they are used in domains like healthcare and finance, where decisions have to be clearly explained [Rudin, 2019]. However, these models sometimes turn out to be less accurate than sophisticated ones. Even more advanced models try to introduce fairness into how they learn explicitly. Such models are called fairness-aware models. For example, adversarial debiasing trains a prediction model along with a second model (an adversary) that tries to guess sensitive information like gender or ethnicity from the predictions. The goal is to make the prediction model "blind" to that sensitive data [Zhang et al., 2018]. Other methods are adaptations of logistic regression or support vector machines specifically designed to avoid unfair outcomes [Zafar et al., 2017]. Collaborative and human-involved systems are another useful line of research. In multi-agent systems, several agents work together and can negotiate to make fair decisions [Powers and Conitzer, 2022]. In situations where the stakes are high, human-involved systems allow people to interrupt and authorize or correct decisions made by the AI [Holstein et al., 2019].

Assessing fairness in MABs is particularly useful in sequential decision-making settings where decisions must be made repeatedly over time and feedback is limited. MABs are therefore particularly well-suited compared to other models for this type of problem. MABs naturally capture the tension between learning (exploring new options) and acting optimally (exploiting known best options). Fairness in this context may involve ensuring that all groups are explored sufficiently or no group is systematically underrepresented during learning. When the system only observes the outcome of the chosen action (not the counterfactuals), MAB is suitable. This is common in real-world systems where only selected individuals receive a treatment or opportunity.

1.4 Modeling Decision-Making: from MAB to CMAB with expert advice

1.4.1 Multi-Armed bandits

Although bandits were briefly introduced in Section 2.3, we now present a more comprehensive explanation. The MAB problem is a classic model in probability theory and machine learning that exemplifies the exploration-exploitation tradeoff. In this problem, a decision-maker must choose among multiple options (arms) with uncertain rewards, aiming to maximize the total reward over time. The challenge lies in deciding whether to exploit known options or explore new ones to discover potentially better rewards. These algorithms are designed to identify the most effective methods of presenting advertisements and social media content that are likely to be engaging and appealing to the user. They are actually optimization algorithms that are also used for many scientific purposes and clinical trials. One could envision many ways to address the exploration-exploitation tradeoff, for instance, prioritizing articles that are more likely to be truthful and less likely to incite rage. In simple MABs, decisions are made without any external context, the learner relies solely on past rewards to refine its strategy. Common algorithms used to solve MAB problems include: *ϵ-greedy*, *Upper Confidence Bound (UCB)*, *Thompson Sampling*.

The efficiency of a MAB algorithm is assessed mainly by a measure called regret. Regret measures

the difference between the reward the algorithm could have obtained by always playing the most profitable arm and the reward actually obtained by the algorithm. The most common example of an MAB problem is a gambler in a casino choosing between multiple slot machines (arms). Each machine has an unknown probability of payout, and the gambler must learn over time which machine offers the highest expected reward while minimizing losses due to poor choices.

1.4.1.1 ϵ - greedy algorithm

The ε-greedy algorithm is a simple yet effective strategy for solving the multi-armed bandit problem by explicitly managing the trade-off between exploration and exploitation [Sutton and Barto, 2018]. At each time step t, the algorithm selects a random arm with probability $\varepsilon \in [0, 1]$, and with probability $1 - \varepsilon$, it selects the arm i with the highest empirical mean reward $\bar{x}_i(t)$. Formally, the selection rule is given by:

$$A_t = \begin{cases} \text{random arm} & \text{with probability } \varepsilon, \\ \arg\max_i \bar{x}_i(t) & \text{with probability } 1 - \varepsilon. \end{cases}$$

Here, $\bar{x}_i(t)$ denotes the average reward obtained from arm i up to time t. The parameter ε controls the degree of exploration: a higher value of ε promotes more exploration, while a lower value favors exploitation. Despite its simplicity, the ε-greedy strategy can achieve good empirical performance and serves as a baseline in many reinforcement learning scenarios. It is useful as a simple baseline or in constrained contexts.

1.4.1.2 Upper Confidence Bound algorithm (UCB)

The Upper Confidence Bound (UCB) algorithm addresses the exploration-exploitation trade-off in the multi-armed bandit problem by selecting, at each time step t, the arm i that maximizes the following expression:

$$\text{UCB}_i(t) = \bar{x}_i(t) + \sqrt{\frac{2 \ln t}{n_i(t)}}$$

Here, $\bar{x}_i(t)$ denotes the empirical mean reward of arm i up to time t, $n_i(t)$ is the number of times arm i has been selected, and $\ln t$ introduces a logarithmic growth in exploration over time. The first term promotes the exploitation of arms with high observed rewards, while the second term encourages the exploration of less frequently chosen arms. This principle ensures that, over time, the algorithm asymptotically identifies the optimal arm with high probability, while maintaining a sub-linear cumulative regret [Auer et al., 2002a]. UCB is a robust choice with good theoretical guarantees. It is a good compromise between exploitation and exploration. However, it performs less well in practice on certain distributions: UCB tends to perform less well in practice when dealing with high-variance distributions (e.g., Gaussian with large variance), non-stationary reward distributions, delayed feedback, or many arms with near-identical expected values.

1.4.1.3 Thompson Sampling algorithm (TS)

In one of the earliest works on stochastic bandit problems, Thomson proposed a natural randomized Bayesian algorithm to minimize regret [Thompson, 1933]. The basic idea is to assume a simple prior distribution on the parameters of the reward distribution of every arm, and at any time step, play an arm according to its posterior probability of being the best arm. This algorithm is a member of the family of randomized probability matching algorithms [Agrawal and Goyal, 2012]. It draws probabilities from a Bayesian distribution. However, it adapts well to uncertainty but is more difficult to implement.

1.4.1.4 EXP3

The EXP3 (Exponential-weight algorithm for Exploration and Exploitation) algorithm is a widely used strategy for solving the adversarial MAB problem. At each round, EXP3 assigns a probability distribution over the arms using exponential weights based on past observed rewards. It then samples an arm according to this distribution and updates the weights of the selected arm using importance-weighted rewards. This allows the algorithm to balance exploration (trying all arms) and exploitation (favoring high-reward arms) in a principled way. EXP3 guarantees a regret bound of $\mathcal{O}(\sqrt{KT \log K})$, where K is the number of arms and T is the number of rounds. It is particularly useful in settings where rewards may be chosen by an adversary or change over time.

1.4.2 Contextual Multi-Armed Bandits

Contextual multi-armed bandits (CMAB) extend this framework by incorporating context or environmental information into the decision-making process. In CMAB, each decision is informed by contextual features, allowing the model to tailor choices based on the current situation. This is particularly useful in dynamic environments where the relevance of options may change with context. Thus, a natural extension to the standard multi-armed bandit is to make decisions with "side information." In this setting, at each time, the agent observes some contextual information, and chooses an arm based not only on the arms' histories, but also on the contextual information CMAB extends the traditional MAB framework by introducing contextual information that influences reward distributions. At each decision point t, the learner observes a context vector x_t (e.g., user preferences, environmental conditions) and selects an arm based on both this context and past experiences. The reward function is no longer fixed per arm but instead depends on the combination of the selected arm and the given context.

The goal in CMAB is to learn a policy:

$$\pi_t : \mathbb{R}^d \to \Delta([K]) \tag{1.1}$$

$\pi_t : \mathbb{R}^d \to \Delta([K])$ denotes a policy at time t that maps a d-dimensional context vector to a probability distribution over K possible arms. Specifically:

- $\mathbb{R}^d$ represents the d-dimensional context space.
- $[K]$ denotes the set of arms, i.e., $\{1, 2, \ldots, K\}$.
- $\Delta([K])$ is the probability simplex over $[K]$, i.e., the set of all probability distributions over the K arms.
- π_t is the policy at time t, which selects actions based on the observed context.

This policy will map contexts to probability distributions over the arms, enabling adaptive decision-making based on available contextual data.

1.4.2.1 Linear Upper Confidence Bound (LinUCB)

LinUCB extends the classical Upper Confidence Bound (UCB) algorithm by incorporating contextual information to make more informed decisions [Li et al., 2010] [Chu et al., 2011].

LinUCB works by assuming that the expected reward of an arm follows a linear model:

$$\mathbb{E}[r_t | x_t, a] = x_t^T \theta_a \tag{1.2}$$

where x_t is the d-dimensional context vector, θ_a is an unknown parameter vector specific to arm a, and r_t is the observed reward. Since θ_a is initially unknown, LinUCB estimates it from past observations using ridge regression. Finally, To select an arm, LinUCB computes an upper confidence bound for each action:

$$\hat{r}_a = x_t^T \hat{\theta}_a + \alpha \sqrt{x_t^T A_a^{-1} x_t} \tag{1.3}$$

where $\hat{\theta}_a$ is the estimated parameter vector, A_a is a covariance matrix tracking feature information, and α controls the exploration-exploitation trade-off. The algorithm selects the arm with the highest upper confidence bound.

1.4.2.2 Contextual Thompson Sampling (CTS)

CTS extends the framework of TS to the CMAB setting[Agrawal and Goyal, 2012]. In contrast to deterministic strategies, CTS incorporates contextual information into the decision-making process by maintaining a posterior distribution over the unknown parameter θ, which models the relationship between context features and expected rewards. This probabilistic approach allows CTS to effectively balance exploration and exploitation while adapting to changes in the environment.

At each time step t, the algorithm samples a parameter estimate from the posterior distribution:

$$\theta_t \sim \text{Posterior}(\theta|\text{past rewards}) \tag{1.4}$$

Given this sampled parameter, CTS selects the arm a that maximizes the expected reward under the sampled model:

$$a_t = \arg\max_{a \in [K]} x_t^T \theta_t \tag{1.5}$$

where:

- $[K]$ is the set of available arms (i.e., $\{1, 2, \ldots, K\}$).
- $x_t \in \mathbb{R}^d$ is the context vector observed at time t.
- $\theta_t \in \mathbb{R}^d$ is a parameter vector sampled from the posterior distribution over model parameters at time t.
- $x_t^T \theta_t$ is the estimated reward for arm a, obtained by taking the inner product between the context and the sampled parameter vector.
- $\arg\max$ selects the arm a_t with the highest estimated reward.

The algorithm balances exploration and exploitation by sampling θ_t from its posterior, which incorporates uncertainty about the reward model. Once the reward is observed, the posterior is updated using Bayesian inference, refining the model's understanding of the reward distribution.

1.4.2.3 Neural Contextual Bandits (Neural UCB / Neural TS)

To be comprehensive, we should also include neural-based algorithms in our discussion. Neural contextual bandits extend traditional contextual bandit models by replacing the typical linear reward model with a deep neural network. Instead of assuming that rewards are linearly dependent on features, these models learn a more flexible function:

$$r = f(x, \theta) + \epsilon, \tag{1.6}$$

where $f(x, \theta)$ is a neural network parameterized by θ, and ϵ represents noise. This approach allows for modeling complex, non-linear relationships between context x and rewards, which is particularly useful in settings where linear assumptions fail to capture the true reward structure.

Neural UCB estimates uncertainty using a confidence bound based on the neural network's predictions. The algorithm constructs an upper confidence bound on the reward for each arm and selects the arm with the highest upper bound. Since neural networks do not provide inherent uncertainty estimates

like Gaussian processes, techniques such as bootstrapped ensembles, last-layer Laplace approximations, or Bayesian linear regression on the final layer are often used to quantify uncertainty.

Neural TS follows the Thompson Sampling framework, where uncertainty is modeled probabilistically. The algorithm samples parameters from a posterior distribution over the neural network's weights and selects actions based on these sampled parameters. Similar to Neural UCB, the challenge lies in estimating uncertainty, which is often handled using variational inference, dropout-based Bayesian approximations, or deep ensembles.

1.4.3 CMAB with expert advice

Bandits with expert advice is a learning framework that combines MAB with expert recommendations. This approach is useful when the agent does not know the optimal action but can rely on one or multiple experts providing potential strategies.The strategy of using the best expert is a good one if that expert is optimal. If not, it is better to use a combination of experts.

In collective decision-making scenarios, multiple agents or experts contribute their judgments toward a common goal. Modeling CDM as a CMAB problem is particularly difficult. This is probably why there are so many algorithms for this. We will mention a few of them below.

1.4.3.1 EXP4: Exponentially Weighted Average Forecaster

EXP4 extends the EXP3 algorithm to the contextual setting by incorporating advice from multiple experts. At each round, it combines expert recommendations to form a probability distribution over actions, balancing exploration and exploitation[Auer et al., 2002b].

1.4.3.2 EXP4.P

EXP4.P is an enhanced version of EXP4 that is more robust in adversarial settings. It modifies the weight update rule to improve performance when expert advice is unreliable or inconsistent.
[Beygelzimer et al., 2011]

1.4.4 Meta-CMAB and oracle-based CMAB

MAML-Bandit[Vuorio et al., 2019] adapts the model-agnostic meta-learning framework to contextual bandits, optimizing for fast policy adaptation. *PEMAB*[Rakelly et al., 2020] incorporates posterior sampling with a hierarchical Bayesian structure to learn useful priors. *Meta-LinUCB*[Kveton et al., 2021] extends linear bandit algorithms using task-shared embeddings for contexts. These methods represent important steps toward developing more sample-efficient and generalizable decision-making systems.

Several contextual bandit algorithms rely on oracles to simplify decision-making or improve learning efficiency. These oracles typically provide predictions based on supervised data or optimization routines. For example, the Epoch-Greedy algorithm [Langford and Zhang, 2008] trains a supervised classifier on logged data and alternates between exploration and exploitation phases. Similarly, the ILOVETOCONBANDITS algorithm [Agarwal et al., 2014] uses a cost-sensitive classification oracle to select policies using importance-weighted loss minimization. These oracle-based methods are especially valuable in large-scale settings, as they reduce contextual bandit learning to well-understood supervised learning problems. Such algorithms are referred to as *oracle-efficient* and often achieve near-optimal regret bounds with practical scalability. In the *exploitation phase*, the oracle is queried to predict the best action given a new context. In *Epoch-Greedy* [Langford and Zhang, 2008], the oracle is retrained at the beginning of each epoch using all collected data so far, and its predictions guide exploitation for the

duration of the epoch. These oracle interactions enable the bandit algorithm to leverage powerful function approximators from supervised learning, while still satisfying exploration/exploitation trade-offs inherent in bandit settings.

Meta-CMAB consists of a framework that addresses expert bias in collective decision-making. Rather than selecting the single best expert, which can be suboptimal due to bias or variability in expertise, the Meta-CMAB algorithm learns an optimal aggregation of expert advice over time. Meta-CMAB leverage prior experience across multiple contextual bandit tasks to accelerate learning in new environments. These approaches apply meta-learning principles to learn shared representations, good initializations, or informative priors, enabling rapid adaptation when faced with a new task. This is particularly useful in non-stationary or few-shot settings, where data is scarce or the environment changes over time.

The core idea is to treat expert predictions as contextual features, and to apply standard CMAB algorithms to learn a policy that maps this meta-context to an arm-selection decision. Here are several variants: Meta-CMAB (FALCON) uses inverse gap weighting to guide exploration under the assumption of realizability; Meta-CMAB (ILTCB) uses an online approximation of ILTCB, maintaining a sparse distribution over policies derived from linear combinations of expert advice; Meta-CMAB (UCB) applies a linear bandit approach with uncertainty-driven exploration based on ridge regression.

Regret bounds can be analyzed for each approach. Meta-CMAB outperforms baseline algorithms such as EXP4-IX and Meta-MAB, particularly when expert predictions are diverse or even negatively correlated. The performance gain stems from the ability to surpass the best individual expert through collective aggregation. This is demonstrated by two misspecification measures: ζ, representing the gap between the best expert and the true reward function, and ε, representing the gap between the best linear combination of experts and the truth. The smaller ε is compared to ζ, the more advantageous aggregation becomes. Moreover special cases involving *homogeneous*, *heterogeneous*, and *polarized* expert populations show that diversity among experts improves the performance of linear aggregation, and that collective intelligence benefits most when experts differ substantially in expertise or exhibit structured disagreement[Abels et al., 2023a].

Overall, the Meta-CMAB framework is theoretically robust and practically effective, enabling higher average reward and lower regret compared to traditional expert-selection methods, especially in realistic, biased, or adversarial settings.

1.4.5 Algorithmic Bias

Machine learning models inherit biases from human data. Algorithmic bias refers to systematic distortions in the results produced by algorithms, often due to biased training data or design choices that reflect human prejudices. These biases can lead to unfair or discriminatory decisions, affecting various areas such as recruitment, justice, and healthcare. Algorithmic bias occurs when the results of an algorithm are not neutral, fair, or impartial, often due to biases present in the training data or introduced during the design of the algorithm. These biases may be unintentional and reflect existing discrimination in society. For example, a resume sorting algorithm may favor male applicants if the training data reflects an overrepresentation of men in previous positions. Similarly, facial recognition systems may have higher error rates for people with dark skin or women if these groups are underrepresented in the training data. The origins of algorithmic bias are manifold: biased training data, algorithm design choices, or the underrepresentation of certain groups in the data.

Fairness in AI is now a major research area, using methods like multi-armed bandits to reduce bias in decision-making. Critics argue that biases might not be errors, but adaptive strategies for real-world decision making [Gigerenzer and Gaissmaier, 2011]. Modern research explores ecological rationality, where heuristics are seen as optimized for specific environments.

1.4.6 How to mitigate algorithmic bias

Algorithmic bias in CMAB settings arises when certain groups systematically receive suboptimal actions due to historical data imbalances, model assumptions, or feedback loops. To mitigate such biases, several strategies have been proposed. Mitigating bias in CMAB requires a careful balance between reward maximization and equitable treatment, and must be aligned with the social and ethical context of the application.

One approach involves incorporating fairness constraints directly into the learning objective, such as demographic parity or equalized odds, which can be enforced during action selection [Joseph et al., 2016].

Another method is to maintain group-specific models or reward estimators, ensuring that under-represented groups are not overshadowed by the majority population [Valko et al., 2021]. Exploration strategies can also be adjusted to guarantee sufficient coverage of all groups, avoiding the exploitation of short-term biases. Moreover, post-hoc reweighting or correction techniques can be applied to logged bandit data to retrospectively evaluate or train fairer policies [Dudík et al., 2011].

MetaCMAB dynamically aggregates expert advice by adjusting weights based on observed predictive performance. This approach not only reduces overall error but also limits the influence of sensitive attributes (e.g., gender, ethnicity) on prediction accuracy[Abels et al., 2023a]. ExpertiseTree goes a step further by modeling localized expertise through adaptive decision trees. It learns context-specific regions in which different experts are more or less reliable, enabling it to isolate and minimize group-specific biases in high-dimensional decision spaces[Abels et al., 2023b]. In particular, ExpertiseTree statistically neutralizes the impact of sensitive variables on prediction errors, offering a principled strategy to improve both accuracy and fairness in biased environments.

1.5 Formulation of Research Questions

The challenge of collective expert decision-making, particularly its susceptibility to bias, has received increasing attention in recent years. Algorithmic biases can profoundly affect the accuracy, reliability, and fairness of aggregated decisions, especially in high-stakes or sensitive contexts. These biases may arise from structural inequalities, cognitive distortions, or even flawed data representations, and they tend to amplify when multiple sources are combined without appropriate safeguards.

In response to these concerns, a growing body of research has focused on developing novel online learning algorithms capable of aggregating expert advice in a principled and adaptive manner. These algorithms aim not only to enhance predictive performance but also to mitigate the distortive effects of bias, whether individual, group-based, or systemic. By dynamically weighting expert inputs based on their past reliability, contextual relevance, or fairness constraints, such approaches offer a promising framework for robust and equitable decision-making in dynamic environments. [Abels et al., 2023a].

In particular, the contextual CMAB framework with expert advice has been used to design algorithms such as MetaCMAB and ExpertiseTrees. These methods dynamically adjust the influence of individual experts based on their observed predictive performance, allowing the system to favor experts whose advice aligns more closely with actual outcomes., enabling improved decision-making in uncertain and heterogeneous environments. By managing the aggregation of expert knowledge in a principled way, these approaches have the potential to enhance both accuracy and fairness, even in settings with systematic biases or evolving expertise.

However, despite these advances, the question of fairness remains largely unaddressed in practice. To date, none of these techniques has been explicitly implemented or evaluated with fairness constraints in mind.

Given this context, several key research questions arise:

1. To what extent do biases impact fairness in Multi-Armed Bandit (MAB), Contextual Multi-Armed Bandit (CMAB), and CMAB with expert advice algorithms?
2. What role do these biases play in the decision-making process within such frameworks?
3. What methods can be employed to effectively mitigate these biases?
4. Are the currently proposed general mitigation strategies sufficient, or is there a need for more tailored approaches in bandit-based learning systems?

1.6 Evaluating Fairness in CMAB: Methods, Metrics, Datasets

As stated above, determining whether an AI system is fair is a challenging task that entails choosing the appropriate definition of fairness and applying it consistently across the system's decision-making. For bandit-based models, this is especially important, since the decision-making process is not only dynamic but also dependent on learning from feedback over time. Fairness for AI can be many different standards, including demographic parity, which requires that different groups come to have equitable consequences equally. Calibration, being interested in the correctness of forecast results per group, is also generally considered [Dwork et al., 2012, Barocas et al., 2019]. In the context of bandits, fairness can also refer to fairness at the arm level, where all arms (e.g., representing different users, services, or groups) have equitable chances of being selected [Joseph et al., 2016, Mehrabi et al., 2021]. There are algorithms that enforce fairness constraints directly in the decision-making process. For instance, the fair contextual bandit algorithm by Joseph et al. [Joseph et al., 2016] ensures that similar groups are handled in the same way during learning. Variants of the UCB and Thompson Sampling algorithms have also been modified to include fairness, allowing for more equitable exploration and exploitation [Joseph et al., 2016]. These methods become more relevant in applications such as personalized healthcare, hiring systems, and recommendation engines, where decisions have real-world consequences for individuals from diverse backgrounds.

When dealing with a typical MAB, which has no contextual data when deciding, fairness is largely concerned with action choice distribution among the arms. If every arm represents a different group, then a fair MAB should not disproportionately ignore some arms. For example, fairness can be quantified by checking how often each arm is selected and ensuring that arms with similar reward distributions are handled equally. This can also be supplemented by computing cumulative regret for each group and comparing their performance with the performance of a fair baseline policy.

When moving from classical multi-armed bandits to *contextual* bandits, fairness evaluation becomes more nuanced. In contextual settings, the algorithm receives a context vector (e.g., user features, environment variables) at each round and uses it to choose an action. This introduces a risk of *contextual discrimination*, where the policy may learn to exploit subtle correlations between sensitive attributes (or their proxies) and expected rewards. As a result, individuals from different groups, despite having similar contexts, might be treated unequally [Joseph et al., 2016, Jabbari et al., 2017].

An additional layer of complexity arises when contextual bandits operate with expert advice, where the algorithm has access to a pool of experts, each providing action recommendations based on the current context. This framework, often used when expert heuristics or policies are available, raises unique fairness challenges. In particular, if the expert pool contains biased or unfair policies, the bandit algorithm may inherit or even amplify these biases, especially during the early learning stages when it is still identifying the best experts.

1.6.1 Existing algorithms for fairness evaluation

The standard evaluation protocol involves simulating an environment with predefined groups, either synthetic or based on real data. Different algorithms, both standard and fairness-aware, are then deployed in

this environment. Performance is assessed using multiple metrics: overall efficiency (e.g., total regret), group-level fairness (e.g., disparity in selection or rewards), and the trade-off between performance and fairness.

Several fairness-aware bandit algorithms have been proposed, such as Fair-UCB, Fair Thomson sampling, and FairLinUCB. These algorithms incorporate explicit fairness constraints during the exploration process to prevent systematic exclusion of specific arms or groups [Jabbari et al., 2017]. Their fairness can be empirically evaluated using synthetic environments (where group assignments are controlled) or through carefully stratified real-world datasets, focusing not on the dataset's fairness per se, but on the algorithm's decision patterns over time.

Fairness evaluation in bandits is not just about outcomes but about the dynamic process of decision-making. A fair bandit algorithm should strive to balance efficiency and equity throughout its learning horizon. Several open-source libraries can support fairness evaluation in practice. For example, Microsoft's Fairlearn offers tools to measure and mitigate bias in decision-making systems, including contextual bandits [Bird et al., 2020]. IBM's AI Fairness 360 toolkit includes a range of metrics and algorithms for bias detection and correction [Bellamy et al., 2018]. Google*What-If* also developed Tool APIs and algorithms to quantify and mitigate bias in AI systems.

1.6.2 Methods

We shall begin by studying the fairness concerns of MAB algorithms without fairness constraints. We then propose to compare the differences between these algorithms and those with fairness constraints, starting with stochastic MABs, then contextual MABs, and finally contextual MABs with expert advice. Before Fair UCB, fairness metrics such as demographic parity or equalized odds could be directly applied to static datasets. Fairness in sequential decision-making, especially in bandit settings, had received little attention, and most multi-armed bandit algorithms were designed purely to minimize cumulative regret, without considering the disparate impact of decisions across individuals or groups.

1.6.2.1 Fair UCB

For bandit-specific research, implementations of fairness-aware bandit algorithms, such as those developed by Joseph et al. [Joseph et al., 2016], could be a starting point. Their *Fair UCB* algorithm addresses fairness concerns in MABs problems. In standard MAB settings, arms with slightly lower estimated rewards can be prematurely ignored, which can lead to unfair treatment, particularly when arms correspond to individuals or demographic groups. Fair UCB mitigates this issue by ensuring that arms with similar expected rewards are selected with similar frequency. Formally, the algorithm enforces a notion of *individual fairness*: if two arms i and j have similar expected rewards, i.e., $|\mu_i - \mu_j| \leq \epsilon$, then their selection probabilities must also be close, such that

$$|\Pr[i \text{ selected}] - \Pr[j \text{ selected}]| \leq \delta,$$

where ϵ and δ are small positive constants. To achieve this, Fair UCB maintains a *plausible set* of optimal arms, those whose upper confidence bounds overlap with the maximum estimated value. It selects uniformly or fairly from this set, ensuring sufficient exploration of all near-optimal arms.

This model was one of the first to propose methods for empirically assessing fairness in bandit algorithms, using selection frequency comparisons between arms and groups. Enforcing fairness in decision-making can come at the cost of higher regret, thus formalizing the fairness-performance trade-off in online learning. Empirical fairness can be evaluated through several methods. First, an *individual fairness test* compares the empirical selection frequencies of arms with similar empirical means; large deviations may signal unfair treatment. Second, the *selection disparity* between arms i and j can be measured as

$$\left| \frac{N_i}{T} - \frac{N_j}{T} \right|,$$

where N_i is the number of times arm i has been selected over T rounds. High disparity between arms with close expected rewards indicates potential unfairness. Third, inspired by *equal opportunity* in supervised learning, one can verify whether arms with high expected rewards have roughly equal selection probabilities across different groups:

$$\Pr[\text{selected} \mid \mu_i \approx \text{high}] \approx \text{equal across groups.}$$

Finally, plotting the cumulative selection count for each arm over time can reveal exclusion of certain arms during early rounds, which is a sign of biased exploration.

1.6.2.2 Fair Thomson sampling

Fair Thompson Sampling is a modification of the classical Thompson Sampling algorithm designed to incorporate fairness constraints across sensitive groups (e.g., age, gender, ethnicity). The goal is to balance exploration and exploitation while ensuring that decisions (e.g., arm selections) do not disproportionately favor one group over others.

In this approach, each group is associated with a separate posterior distribution for each arm. At each round, a sample is drawn from the posterior for each arm and group, and an arm is selected in a way that maximizes expected reward *subject to a fairness constraint*. One commonly used constraint is demographic parity, which ensures that each group has an approximately equal chance of receiving a favorable decision.

To enforce fairness, additional mechanisms such as tracking the number of selections per group, applying constraints on arm selection probabilities, or using rejection sampling are introduced. The algorithm then updates the posteriors based on observed rewards, similarly to classical Thompson Sampling [Jabbari et al., 2017].

This algorithm allows practitioners to trade off between reward maximization and fairness guarantees in online decision-making settings.

1.6.2.3 LinUCB with fairness constraints

While the standard LinUCB selects arms based on an upper confidence bound of the expected reward, Fair LinUCB introduces constraints to ensure that the selection process remains equitable across sensitive groups (e.g., gender, ethnicity, or age).

In each round, the algorithm computes the UCB score for each arm using contextual features, just as in classical LinUCB. However, it incorporates a fairness constraint, typically demographic parity, by enforcing that the selection probabilities for different groups remain balanced over time. This is often done by modifying the arm-selection rule to limit disparities in selection frequency or by introducing a fairness-aware regularization term in the decision criterion.

By doing so, Fair LinUCB ensures that no group is consistently favored or neglected, even if their contextual features might lead to slightly different expected rewards. The algorithm strikes a balance between maximizing overall utility and maintaining equitable treatment across groups [Joseph et al., 2016].

1.6.2.4 MetaCMAB

Meta-CMAB is a meta-learning algorithm that addresses biased or heterogeneous expert advice in contextual multi-armed bandit (CMAB) settings. Rather than selecting the best expert, Meta-CMAB learns an optimal combination of expert predictions. At each decision step, experts observe the context and provide advice (e.g., predicted rewards for each action). These advice vectors are then used as features in a secondary CMAB, termed the Meta-CMAB. The algorithm learns a policy mapping expert advice to actions, optimizing a linear model that predicts expected rewards from expert opinions.

1.6.2.5 ExpertiseTree

ExpertiseTree is an algorithm designed to adaptively identify and leverage expert knowledge in a context-sensitive way. Unlike MetaCMAB, which globally adjusts expert weights, ExpertiseTree builds a decision tree over the context space to localize which experts perform best under specific contextual conditions. Its goal is to address the fact that experts are often not equally reliable across all contexts, some may be more accurate in certain regions of the input space, while others are better elsewhere. This context-dependence is especially important when group-sensitive biases are present (e.g., some experts might systematically underperform for a particular ethnic or gender group).

1.6.2.6 Microsoft's Fairlearn

Microsoft's Fairlearn is a Python library designed to assess and mitigate unfairness in supervised machine learning models, particularly in classification and regression tasks. It offers tools to evaluate disparities across different demographic groups and to implement fairness-aware algorithms. However, Fairlearn is not designed for real-time integration with online bandit algorithms.

1.6.2.7 IBM's AI Fairness 360 toolkit

IBM's AI Fairness 360 toolkit is not tailored for online bandit algorithm training. However, it could serve for evaluating and mitigating fairness issues in the outcomes of MAB algorithms using offline data. For real-time fairness integration, additional tools and algorithms specifically designed for fair bandit learning should be used.

1.6.2.8 Google*What-If*

Google's What-If Tool (WIT) is designed to analyze and visualize machine learning models without requiring code, offering features to assess model performance and fairness across different subgroups. However, to evaluate the fairness of multi-armed bandit (MAB) algorithms, offline settings is mandatory.

1.6.3 Metrics

1.6.3.1 Average reward

The *average reward* is a standard performance metric used to evaluate the effectiveness of algorithms in multi-armed bandit settings. It measures the mean of the rewards obtained over a sequence of decisions made by the algorithm. Formally, if r_t denotes the reward received at time step t, and T is the total number of rounds, the average reward is defined as:

$$\text{Average Reward} = \frac{1}{T}\sum_{t=1}^{T} r_t$$

This metric reflects how well an algorithm performs in maximizing immediate returns, without explicitly accounting for fairness or long-term regret. In fairness-sensitive settings, the average reward may also be computed separately for different subgroups (e.g., based on sensitive attributes like ethnicity or gender), which allows for the analysis of disparities in outcomes across groups.

1.6.3.2 Action Frequency

Action frequency refers to the proportion of times each arm (or action) is selected during the learning process. It is often used to assess whether certain arms or subgroups are favored or neglected by the algorithm. Formally, for an arm a_i, its frequency over T rounds is:

$$\text{Frequency}(a_i) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}(a_t = a_i)$$

where $\mathbb{I}$ is the indicator function. This metric is especially useful for analyzing fairness across groups, as it reveals selection imbalances.

1.6.3.3 Fairness Gap

The *fairness gap* measures the disparity in performance metrics, such as average reward or regret, across sensitive groups. For example, given two groups G_1 and G_2, the fairness gap in average reward is defined as:

$$\text{Fairness Gap} = \left|\text{Average Reward}_{G_1} - \text{Average Reward}_{G_2}\right|$$

A smaller fairness gap indicates more equitable treatment of groups. This metric can be extended to more than two groups and applied to other indicators like regret or action frequency.

1.6.3.4 Regret

Regret quantifies the cumulative difference between the reward obtained by the algorithm and the reward that would have been obtained by always selecting the optimal action in hindsight. Let a_t be the action taken at time t, and a_t^* the optimal action. If $r(a)$ denotes the expected reward of action a, then the cumulative regret over T rounds is given by:

$$\text{Regret} = \sum_{t=1}^{T} [r(a_t^*) - r(a_t)]$$

Lower regret indicates that the algorithm has effectively learned to make near-optimal decisions over time. Different types of regret have recently been proposed [Abels et al., 2023a]

1.6.3.5 Contextual Regret

The regret of a contextual bandit policy π over T rounds is defined as the difference between the cumulative expected reward of an optimal policy and that obtained by π:

$$R_T = \sum_{t=1}^{T} \left(\max_{k\in[K]} f(k, \mathbf{x}_t) - f(k_t, \mathbf{x}_t) \right)$$

where $\mathbf{x}_t$ is the context at time t, $f(k, \mathbf{x}_t)$ is the true reward function, and k_t is the arm selected by the algorithm.

1.6.3.6 Expert Regret

Regret can also be computed with respect to the best expert in hindsight:

$$R_T^{\text{expert}} = \max_{n\in[N]} \sum_{t=1}^{T} \left(f(k_t^n, \mathbf{x}_t) - f(k_t, \mathbf{x}_t)\right)$$

where $k_t^n = \arg\max_k \tilde{f}_n(k, \mathbf{x}_t)$ is the arm suggested by expert n at time t, and $\tilde{f}_n$ is that expert's reward estimate function.

1.6.3.7 Expert Misspecification ζ

The misspecification ζ quantifies the worst-case discrepancy between the best expert's advice and the true reward function:

$$\zeta = \min_{n \in [N]} \max_{t \in [T], k \in [K]} \left| f(k, \mathbf{x}_t) - \tilde{f}_n(k, \mathbf{x}_t) \right|$$

A higher ζ indicates poor alignment between the best expert and the ground truth.

1.6.3.8 Linear Misspecification ε

This measures the best achievable approximation error from any linear combination of expert predictions:

$$\varepsilon = \min_{\boldsymbol{\theta} \in \mathbb{R}^N} \max_{t \in [T], k \in [K]} \left| f(k, \mathbf{x}_t) - \langle \boldsymbol{\theta}, \tilde{\mathbf{f}}_{k,t} \rangle \right|$$

where $\tilde{\mathbf{f}}_{k,t} = (\tilde{f}_1(k, \mathbf{x}_t), \ldots, \tilde{f}_N(k, \mathbf{x}_t))$ is the vector of predictions for arm k at time t from all N experts. When $\varepsilon \ll \zeta$, it indicates that aggregating advice leads to a much better approximation than any single expert can achieve.

These metrics enable evaluation of both algorithmic behavior and expert utility in biased settings. In particular, they support the use of Meta-CMAB in collective decision-making scenarios.

1.6.4 Datasets

We propose to test fair and unfair algorithms on the following databases:

1.6.4.1 A synthetic database

Inspired by the framework proposed by [Joseph et al., 2016]. The core idea is to model an environment where different sensitive groups (such as groups A and B) interact with a sequential decision-making system. At each time step, the system selects one of K available actions (or arms) and receives a stochastic reward in return.

In a simulation, each group is associated with a specific reward distribution for each arm. For instance, group A may have higher probabilities of success for certain arms than group B, and vice versa. This asymmetry reflects inherent group-specific preferences or structural inequalities, making it possible to analyze how decision-making algorithms behave under such conditions.

The data generation process unfolds as follows: at each round, a group is randomly sampled according to a fixed distribution (e.g., a 50-50 split between groups A and B). The algorithm then selects an arm, either at random or following a specific strategy such as ε-greedy or Thompson Sampling. The reward is drawn from a Bernoulli distribution whose parameter depends on both the selected group and arm.

Each interaction is recorded as a tuple consisting of the round number, the group identifier, the chosen arm, and the resulting reward. The simulated dataset provides a controlled setting for evaluating algorithmic performance in terms of cumulative regret. We will compare the regret between fair and non fair bandit algorithms.

1.6.4.2 Swiss Chocolate Example (FairBandits repo)

The Swiss Chocolate Example, featured in the FairBandits repository, is a synthetic benchmark designed to illustrate fairness challenges in bandit algorithms under controlled conditions. It simulates a simplified decision-making scenario where two demographic groups (e.g., Group A and Group B) are associated with different success probabilities for each arm (representing different options or actions, such as recommending chocolate types). This setup allows precise manipulation of reward distributions

to induce or correct group-level disparities. In the CMAB framework, the group membership serves as the context, and each arm corresponds to a decision alternative. The reward reflects binary feedback (e.g., success/failure). This example is particularly useful for testing the behavior of fair versus unfair bandit algorithms in a clean, interpretable environment.

1.6.4.3 A health database: MIMIC-III Clinical database

The Medical Information Mart for Intensive Care (MIMIC) is a large, publicly available dataset developed by the MIT Laboratory for Computational Physiology, in collaboration with the Beth Israel Deaconess Medical Center. The two main versions, MIMIC-III contain de-identified electronic health records from over 40,000 intensive care unit (ICU) admissions, covering periods from 2001 to 2012.

The input data will encompass demographic attributes such as age, sex, ethnicity, weight, and height; facilitating fairness assessments across different patient groups. Clinical context will be provided by vital signs, including heart rate, respiratory rate, blood pressure, body temperature, and oxygen saturation, alongside laboratory test results serving as predictive features. The set of actions (arms) will represent various clinical decisions, such as the administration of different treatments. The reward function will be defined based on 30-day survival outcomes and/or the trajectory of lactate levels. Incorporating sensitive group identifiers will enable the evaluation and comparison of how different bandit algorithms perform across diverse patient subpopulations.

Patient features such as age, vital signs, and laboratory results can form the context, while arms correspond to treatment choices (e.g., fluid resuscitation, vasopressors, or mechanical ventilation), and rewards are defined by patient outcomes such as survival, readmission, or length of stay.

Although access to MIMIC is restricted and requires both a data use agreement and documented training in research ethics, authorization has already been requested and the eligibility criteria are, in principle, fulfilled. In particular, formal ethics training was completed during prior undergraduate medical studies, including the certification required for access.

1.6.4.4 COMPAS Recidivism Dataset

The COMPAS Recidivism dataset, collected and published by ProPublica, contains criminal justice data used to assess the likelihood of recidivism among defendants in Broward County, Florida. It includes demographic information (e.g., age, gender, race), criminal history, and risk scores generated by the proprietary COMPAS algorithm. This dataset is particularly relevant for fairness studies, as prior analyses have demonstrated significant racial disparities in the risk scores assigned to Black and White defendants, despite similar ground-truth outcomes. In the context of CMAB, the dataset can be used to model decision-making processes (e.g., bail or sentencing decisions) where the context is the defendant's profile, and the arms represent potential judicial outcomes. The goal is to evaluate whether algorithmic policies lead to equitable treatment across sensitive groups.

1.6.4.5 Student Performance Dataset

The Student Performance dataset, available from the UCI Machine Learning Repository, contains academic and demographic information on Portuguese secondary school students. It includes features such as gender, age, school, parental education, address type (urban or rural), and prior academic performance. The dataset is commonly used to predict students' final grades, but it is also suitable for fairness analysis. In the context of CMAB, the student profile can be treated as the context, and different educational interventions (e.g., tutoring, resource allocation) can be modeled as arms. The reward corresponds to improvements in student performance. This dataset allows researchers to examine whether algorithmic policies equitably allocate educational support across groups defined by sensitive attributes such as gender or socioeconomic background.

1.6.4.6 MovieLens 1M Dataset

The MovieLens 1M dataset, provided by the GroupLens research lab, contains 1 million movie ratings from over 6,000 users and includes demographic attributes such as age, gender, and occupation. Each user is associated with a history of movie ratings, making it a widely used benchmark for recommendation systems. In the context of CMAB, user profiles form the context, and the arms correspond to movie recommendations. The reward is typically derived from user ratings or engagement. This dataset enables fairness-aware evaluation of recommender policies by assessing whether certain groups (e.g., by gender or age) receive systematically different recommendations, potentially leading to biased user experiences or unequal access to content diversity.

1.6.4.7 Adult Income dataset

The Adult Income dataset, sourced from the UCI Machine Learning Repository, contains demographic and employment-related information on over 48,000 individuals from the U.S. census. Each record includes features such as age, education, occupation, race, and gender, with the goal of predicting whether a person earns more than $50,000 per year. This dataset has become a standard benchmark for fairness research due to well-documented disparities in prediction outcomes across sensitive groups. In a CMAB setting, the individual's profile can be used as the context, while the arms represent different decision options, such as targeting for job offers or credit approval. The reward may reflect predicted income or decision success. This dataset enables the evaluation of whether CMAB policies make equitable decisions across subgroups defined by gender or race. The context includes socio-demographic attributes, arms may represent resource allocation decisions (e.g., job offers or training programs), and the reward can be the predicted or observed income level.

Chapter 2

Objectives

The objective of this work is to evaluate and compare the fairness of various algorithms developed to address the CMAB problem, with the aim of identifying which approaches promote more equitable decision-making across different groups or contexts. It is important to clarify that CMAB refers to a problem setting rather than a specific algorithm or model. The focus of our fairness assessment lies in the algorithms designed to solve this problem. To this end, we propose a comparative analysis of three types of bandit algorithms: standard multi-armed bandits (MAB), contextual MAB, and contextual MAB with expert advice. These algorithms will be evaluated on several distinct datasets. Fairness indicators and algorithms' performance will be compared based on whether a fairness-aware or fairness-agnostic variant is used.

In this study, we will follow a standard evaluation pipeline to assess fairness, regardless of the model type. We will start by identifying relevant sensitive attributes, such as ethnicity, gender, or age, within the data sets. A fairness criterion will then be selected based on the specific needs of the application domain. During experimentation, we will log each model's actions and predictions along with the corresponding group identities and observed rewards. These logs will be analyzed to compute group-level metrics, including average reward, action frequency, and cumulative regret. To track how fairness evolves over time, we will generate visualizations of these metrics throughout the learning process. Finally, we will assess each algorithm's fairness by comparing its behavior to that of a fairness-aware baseline.

Moreover, our work will try to evaluate fairness not only globally, but also *per time step*, highlighting temporal disparities between groups; analyze *group-specific regret*, capturing how exploration policies may disproportionately disadvantage certain subpopulations during learning.

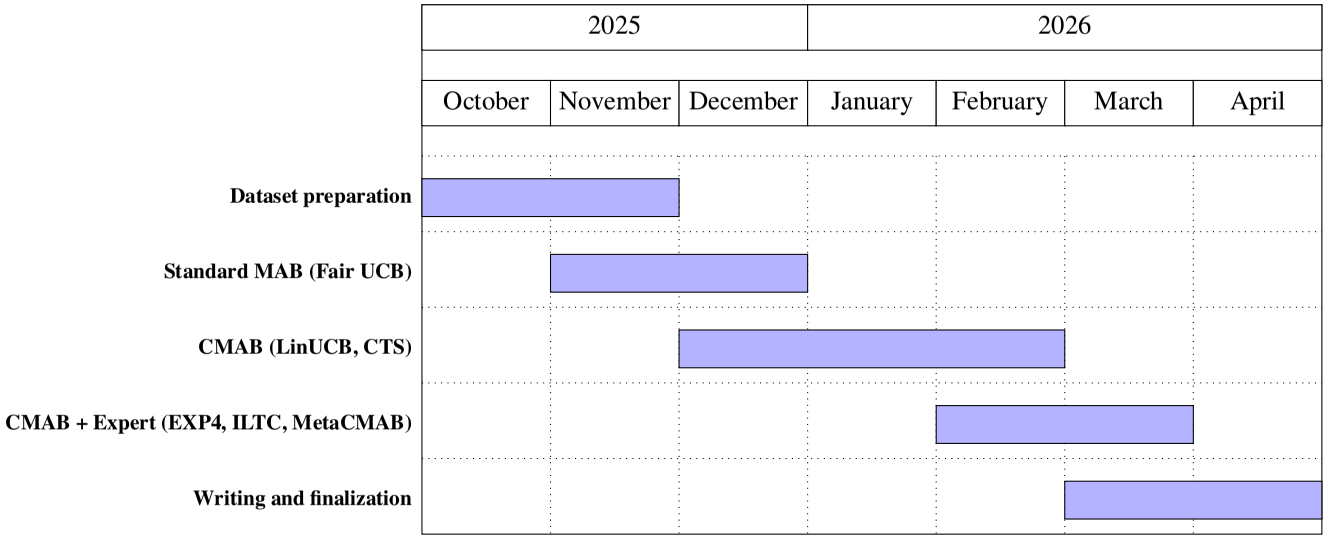
Another objective of this thesis will be to examine how fairness can be actively promoted, or at least its degradation limited through algorithmic design and selection. Accordingly, we shall explore whether integrating fairness-aware strategies can mitigate unfairness in the learning process. This approach will aim to go beyond passive observation and could contribute to the growing literature on fair decision-making under uncertainty.

We propose the following pipeline (see Table 2.1 and Appendix 2): we will start by preparing the datasets, then we will analyze and compare the different types of bandits over the next six months, and we plan to spend a whole month on writing. We are using buffer periods for unforeseen events, which are bound to arise.

Duration	**Problem**	**Model**	**Evaluation**
1 month	Standard MAB	Fair UCB	Averaged reward, regret
2 months	CMAB	LinUCB CTS	Contextual regret
3 months	CMAB + expert	EXP4 ILTC MetaCMAB	Expert regret, expert misspecification, linear misspecification

Table 2.1: Project timeline and evaluation criteria for different bandit settings. **MAB**: Multi-Armed Bandit; **CMAB**: Contextual MAB; **CTS**: Contextual Thompson Sampling; **ILTC**: ILOVETOCONBANDITS. **EXP4**: Exponential Weights for Exploration and Exploitation. **MetaCMAB**: Meta-learning for CMAB with expert advice.

Appendix A – Vertical Gantt Chart: Thesis Timeline



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